

Steering through the storm: Environmental uncertainty and delivery performance

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Funding information

UK Research and Innovation, Grant/Award Number: EP/V011855/1; National Natural Science Foundation of China, Grant/Award Numbers: 72103029, 72371055, 71971041; Major Program of National Social Science Foundation of China, Grant/Award Number: 20&ZD084; Key Program of the National Natural Science Foundation of Sichuan Province, Grant/Award Number: 2024NSFSC0025

Handling Editors: Stephen M. Disney, Jan Holmstrom, Benn Lawson, Frits K. Pil, and Christopher Tang

Abstract

We investigate the impact of Typhoon Lekima, a source of environmental uncertainty, on delivery performance, utilizing data from RiRiShun Logistics. We apply a staggered difference-in-differences model across low-complexity and high-complexity supply chains. The analysis uncovers distinct dynamics throughout the typhoon's forecast, active, and recovery phases. Notably, the active phase experiences less disruption in delivery performance compared to the forecast phase, owing to RiRiShun's wait-and-see policy. This approach leads to order accumulation but subsequently enables more effective resource allocation. In the recovery phase, high-complexity supply chains demonstrate significant improvements, surpassing pre-disaster performance levels. The study further emphasizes the critical role of two response mechanisms in managing high-complexity supply chains. Increasing throughput efficiencies at destination centers and implementing transfer centers prove effective in enhancing delivery performance during both the forecast and active phases. However, the simultaneous application of these response mechanisms during the forecast phase unexpectedly reduces performance.

KEYWORDS

contingency planning, delivery performance, environmental uncertainty, response mechanisms, supply chain complexity

Highlights

- Crises offer a unique opportunity to implement long-term operational improvements that may not be evident during stable periods.
- Deliberately holding back on deliveries during the forecast period of disruption can enable more effective resource deployment during the active phase, leading to a better, more resilient recovery that exceeds pre-disruption performance.
- Using transfer centres as intermediary nodes and adjusting throughput efficiencies at destination centres can enhance resilience, but simultaneously applying both strategies can be counterproductive.

1 | INTRODUCTION

Disruptions in supply chain networks are a well-acknowledged reality among supply chain professionals, with delivery delays remaining a predominant concern (Economist, 2021). The challenge lies in understanding the capabilities of supply chain systems and formulating effective contingency plans to mitigate the impact of external shocks. To investigate the role of robust contingency planning amidst environmental uncertainty, we examine the delivery performance within the expansive distribution logistics network of RiRiShun Logistics. RiRiShun Logistics is a subsidiary of Haier and a leading entity in home appliance delivery in China. Our research capitalizes on the unique opportunity presented by Typhoon Lekima to draw a causal link between environmental uncertainty and delivery performance within the framework of a natural experiment.

Our study spans from July to September 2019, a crucial timeframe for assessing the impact of Typhoon Lekima. The typhoon, by its very nature, introduces variability in disruption (i.e., also known as exogenous shock¹), serving as an archetypal random event that affects delivery performance. This variability is contingent upon the extent of the typhoon's impact, which we interpret as *environmental uncertainty*. Given the cyclical nature of typhoons, encompassing a build-up and landfall pattern, we delineate the impact on delivery performance into four discernible stages of environmental uncertainty: the *pre-disaster* phase, the *forecast* phase, the *active* phase, and the subsequent *recovery* phase.

The *pre-disaster* phase serves as a benchmark in our analysis, representing RiRiShun's operational efficiency under typical conditions. As we enter the *forecast* phase, which begins when the firm receives an alert about the imminent threat of the typhoon, the level of environmental uncertainty significantly escalates. The subsequent *active* phase marks the typhoon's landfall and direct impact. Finally, in the *recovery* phase when the environmental uncertainty is resolving, the firm's recovery depends on its supply chain resiliency. Firms like RiRiShun need to assess the swiftness and efficacy with which normal delivery operations are reinstated, understanding the long-term implications of the disaster on supply chain operations and discerning the lessons learned for future contingency planning. Therefore, utilizing the natural experiment and a staggered difference-in-differences (DiD) methodology, we gain insights into the impact of environmental uncertainty, triggered by the typhoon's external shock, on the delivery performance of consumer appliances.

In addition to establishing a causal relationship between environmental uncertainty and delivery performance,

we examine its effect on supply chain complexity. We distinguish between two levels of supply chain complexity: low and high. A low-complexity supply chain involves deliveries that start and end within the same city. In contrast, a high-complexity supply chain deals with deliveries that begin in a particular city and distribute to others, necessitating coordination across a more complex inter-city network. While low-complexity supply chains are streamlined due to the absence of inter-city logistics, leading to minimal disruptions and lower issue resolution, high-complexity supply chains necessitate extensive coordination due to their susceptibility to disruptions.

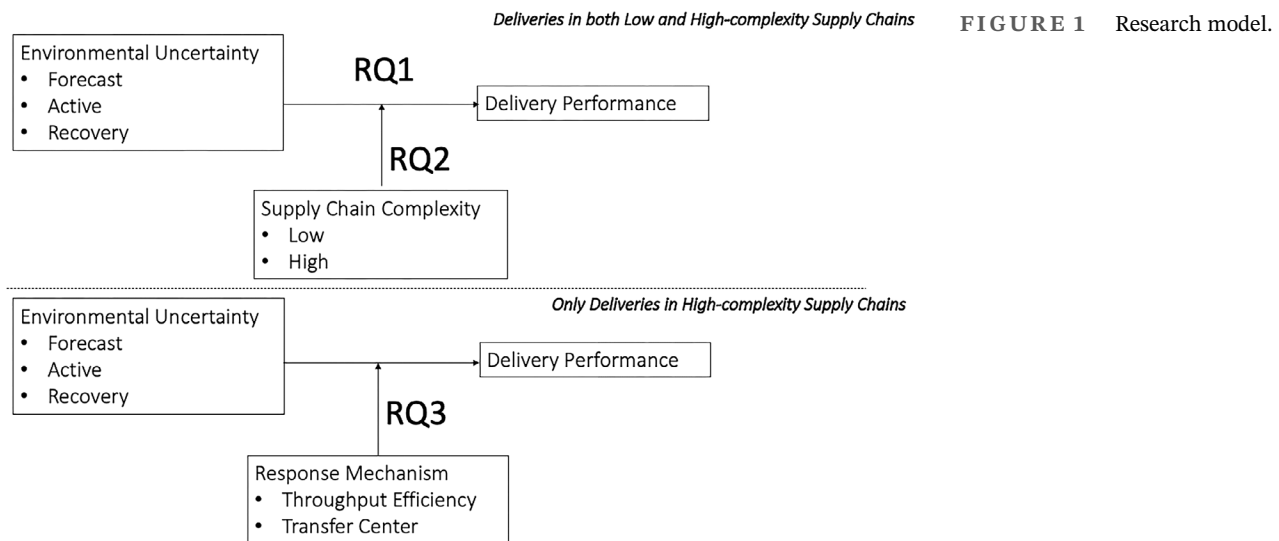
The notion of complexity stems from managing numerous nodes and links, especially during critical inbound and outbound operations between logistical functions. This approach aligns with established literature on supply chain complexity, which assesses complexity based on the network's extent of connections and logistical challenges (see Lu & Shang, 2017; Vachon & Klassen, 2002). Our classification is further validated through semi-structured interviews with RiRiShun's logistics personnel,² highlighting the differences in operational burden between within-city and inter-city deliveries. Specifically, inter-city deliveries are marked by greater logistical scale, resource allocation, and coordination efforts, making them more complex than within-city deliveries, which are generally more straightforward and adaptable to changes.

Focusing on the high-complexity supply chains, we investigate two response mechanisms available to RiRiShun Logistics to mitigate the adverse effects of environmental uncertainty. The first is addressing throughput efficiencies at the destination centers. The second is the strategic use of transfer centers as load optimizing tools. Our analysis is structured around the following research questions:

RQ1. How does environmental uncertainty at different phases of the typhoon impact delivery performance?

RQ2. Does environmental uncertainty impact delivery performance in low-complexity supply chains differently than in high-complexity supply chains across the phases?

RQ3. In a high-complexity supply chain, how do enhancing throughput efficiency at destination centers and utilizing transfer centers mitigate the effects of environmental uncertainty across the phases?



Our research model in Figure 1 outlines the relationships expressed by the three research questions.

2 | PROBLEM SETTING AND DATA PROCESSING

Our study investigates the impact of environmental uncertainty on delivery performance, utilizing the natural experiment offered by Typhoon Lekima. In August 2019, Typhoon Lekima, the most intense typhoon to hit Mainland China since 1961, caused unprecedented damage, with nationwide costs surpassing CNY 53.7 billion.³ We sourced historical data on Typhoon Lekima from the China Meteorological Administration's typhoon monitoring website,⁴ collecting detailed information on the typhoon's pathway, warnings, and forecasts. Typhoon Lekima swept across Mainland China from August 10th to 13th, 2019, with the initial forecast of its landing announced on August 7th, 2019. This timeframe is instrumental in capturing the various phases of the typhoon's impact: from the pre-disaster and initial forecasting phases, through the peak of its influence, and into the recovery period, thereby shedding light on its influence on delivery performance. Figure 2 depicts Lekima's tracking route and impact strength in Mainland China.⁵

The operational-level tracking data for our analysis, focusing on delivery performance, was sourced from the 2021 Manufacturing & Service Operations Management (MSOM) Data-Driven Research Competition,⁶ courtesy of RiRiShun Logistics (Guo et al., 2024). As a subsidiary of the Haier group, RiRiShun is recognized for its specialized logistics services in the home appliance sector.⁷ The comprehensive dataset detailed information on delivery appointments and processes for over 14 million orders,

covering more than 18,000 SKUs, all managed through 103 distribution centers between October 2018 and September 2019.

A key challenge in our study is to identify which distribution centers were affected by Typhoon Lekima. This task involves deciphering the actual delivery routes of the orders and locations of distribution centers, which are anonymized in the RiRiShun dataset. To address this issue, we employed a creative solution. We tapped into RiRiShun's web mapping API⁸ to obtain the exact *pin drop* locations where RiRiShun operates. Using these precise geographical points, we then constructed a comprehensive matrix that measures the travel distances between pairs of locations. This matrix was developed by integrating the latitude and longitude data of these points with the navigation functionalities provided by both Gaode and Baidu mapping services. This approach allowed us to effectively estimate the travel distances and routes.

Building on our earlier efforts, we took the next crucial step of validating the accuracy of our travel distance matrix. This involved a thorough comparison of the distance calculations obtained from both Gaode and Baidu mapping services. Once we established that these calculations were consistent with each other, we proceeded to align our matrix with the anonymized distance matrix provided by RiRiShun. This careful cross-referencing enabled us to accurately identify the city information of 99 distribution centers out of the 103. The distribution centers in cities affected by Typhoon Lekima in our sample are shown in Table 1.

This allows us to robustly quantify the impact of environmental uncertainty on delivery performance, accounting for the staggered nature of the typhoon's impact across different locations and times. As each order only

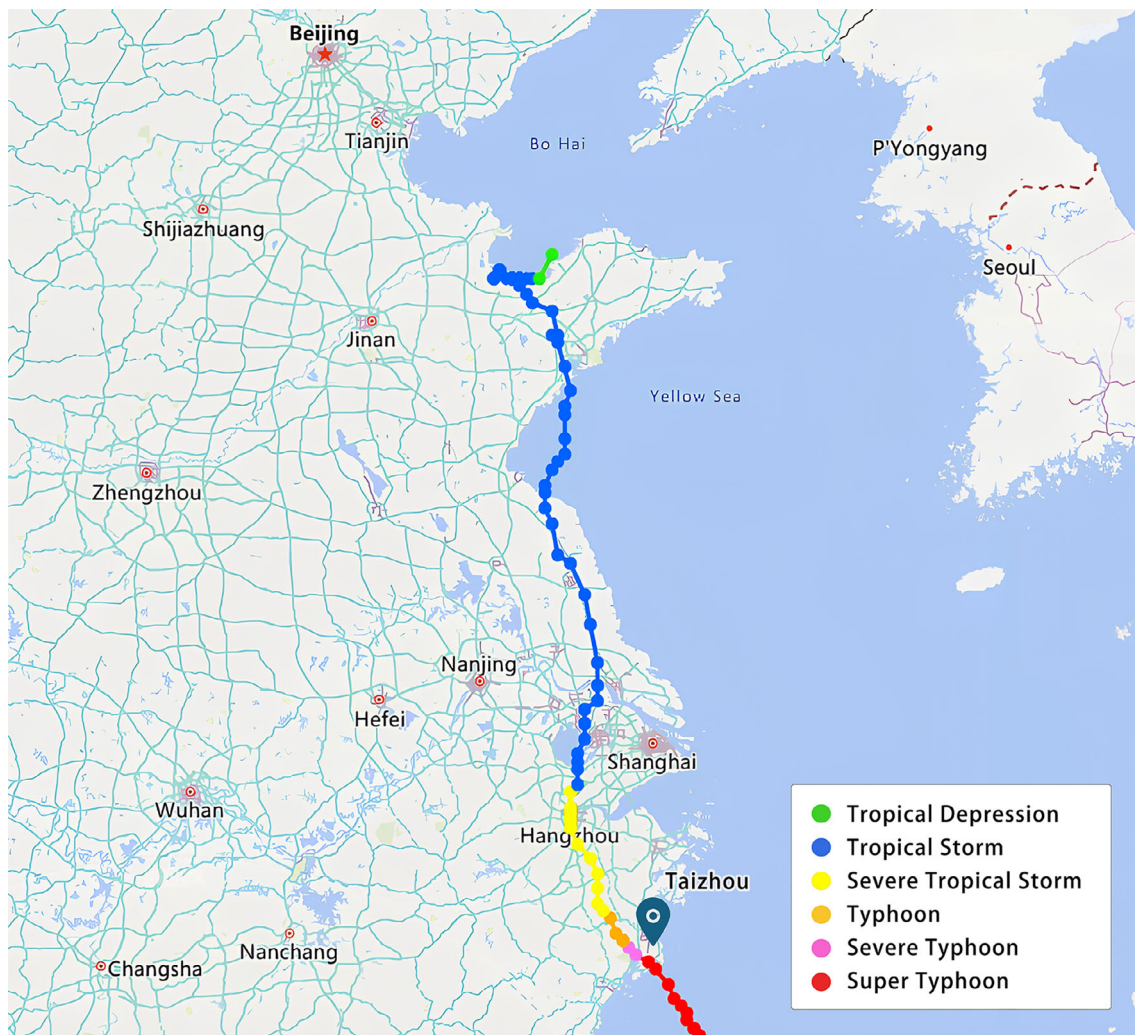


FIGURE 2 Typhoon Lekima path and impact strength in Mainland China.

TABLE 1 Typhoon Lekima: Forecast and active stages for distribution centers in affected cities.

Forecast stage	Active stage	Distribution centers in affected cities
7-Aug-19 to 9-Aug-19	10-Aug-19 to 13-Aug-19	Taizhou
8-Aug-19 to 9-Aug-19	10-Aug-19 to 13-Aug-19	Wenzhou, Suzhou
8-Aug-19 to 10-Aug-19	11-Aug-19 to 13-Aug-19	Wuxi
9-Aug-19 to 9-Aug-19	10-Aug-19 to 13-Aug-19	Hangzhou, Huzhou, Jinhua
9-Aug-19 to 10-Aug-19	11-Aug-19 to 13-Aug-19	Nantong, Yancheng
10-Aug-19 to 10-Aug-19	11-Aug-19 to 13-Aug-19	Qingdao, Weifang

Note: Data are compiled from the China Meteorological Administration website <http://www.cneb.gov.cn> and the China News Service website <https://www.chinanews.com.cn/>.

appears once in the RiRiShun Logistics data, we sort the orders by their SKUs and destination centers to construct a panel dataset at SKU-city level for the empirical analysis. The dataset consists of 458,357 observations from July 2019 to September 2019.

3 | VARIABLES

We now describe the variables and their corresponding operational definitions used in this study.

3.1 | Delivery performance

Delivery performance is critical in supply chain contracts (Dai et al., 2016), which points to the reliability (Handfield & Pannesi, 1992) and quality of the service provider (Shang & Liu, 2011). Research on delivery

performance identifies broadly four dimensions, which are on-time delivery rate, early delivery inaccuracy, late delivery inaccuracy, and delivery speed (Peng & Lu, 2017). Our study focuses on the delivery performance of on-time delivery rate ($Promised_{ijt}$), which is the dependent variable, because it is a core element in service level agreements in logistic operations. For instance, logistics companies such as DHL publish their late delivery credit scale related to compensation for deliveries beyond the promised time.⁹

To ensure consistency and robustness of our results, we use two measures for the dependent variable $Promised_{ijt}$, that is, $PromisedOn_{ijt}$ and $PromisedBy_{ijt}$. The first, $PromisedOn_{ijt}$, is defined as the percentage of orders of SKU i due for destination city j , which are placed at time t , and arrive on their promised delivery dates. The second, $PromisedBy_{ijt}$, is defined as the percentage of orders of SKU i due for destination city j , which are placed at time t , and arrive by their promised delivery dates. We show in Appendix A that promised delivery dates are critical order information communicated to consumers in major platforms.

We include another dependent variable *Fulfillment Time* ($FTime_{ijt}$) to conduct the investigation in Section 4.2. This variable captures the average duration, in hours, from when a customer places an order to when the delivery is received, aggregated at the SKU-city level.

3.2 | Treatment group by Lekima phase

We use $TreatForecast_{ijt}$, $TreatActive_{ijt}$, and $TreatRecovery_{ijt}$ to measure the treatment groups affected by Lekima, where $TreatPre_{ijt}$ is the omitted reference benchmark category. $TreatPre_{ijt} = 1$ if destination city j for SKU i is among the affected cities when day t is within 1 week before the forecast stage of Lekima, and zero otherwise. We define $TreatForecast_{ijt} = 1$ if destination city j for SKU i is among the affected cities when day t is within the forecast stage of Lekima as shown in Table 1, otherwise $TreatForecast_{ijt}$ is equal to zero. $TreatActive_{ijt} = 1$ if destination city j for SKU i is among the affected cities when day t is during the active stage of Lekima, and $TreatRecovery_{ijt} = 1$ if destination city j for SKU i is among the affected cities when day t is within 1 week after the end of Lekima, and zero otherwise.

3.3 | Supply chain complexity

We use the indicator $HighComplex_{ijt}$, which is set to one for deliveries that span multiple cities (high-complexity), and zero for those confined to a single city (low-complexity).

3.4 | Response mechanisms

We examine two distinct response mechanisms, which offer contrasting approaches for RiRiShun to mitigate disruption. The first response mechanism evaluates the flow efficiency of RiRiShun's destination centers. Therefore, we measure *Throughput Efficiency* (TE_{ijt}) as the deliveries outflow to deliveries inflow for destination city j for SKU i at day t . A higher TE_{ijt} value indicates greater flow efficiency. In contrast, the second mechanism is using transfer centers as load optimizing tools to enhance the network's adaptability and resilience (Campbell, 1996; Campbell et al., 2005; Lien et al., 2011). Hence, we use $Transfer_{ijt}$ as a load optimizing tool and specifically account for the percentage of orders for SKU i bound for destination city j ordered on day t that passes through at least one transfer center.

3.5 | Control variables

A primary concern is that several of the independent variables may be endogenous, that is, controlled, at least in part, by the order, SKU, and city. Since we do not observe all factors that affect delivery decisions, the endogenous variables of $PromisedOn_{ijt}$ and $PromisedBy_{ijt}$ can be correlated with the error term, ε_{ijt} . In such a situation, ordinary least squares (OLS) can lead to biased estimates. We seek to mitigate this endogeneity bias by including SKU-city fixed effects η_{ij} , day fixed effects λ_t and the following vector of covariates X : the actual log-transformed travel $Distance_{ijt}$ (Chen et al., 2015; Corsten et al., 2011), measured in average kilometers between the origin and destination center for SKU i due for destination city j on day t ; $ProductQuantity_{ijt}$, which is the average number of products of orders for SKU i due for destination city j on day t ; $LastMileHub_{ijt}$ which is the percentage of orders distributed through a last-mile hub for SKU i due for destination city j on day t ; and $Congestion_{jt}$ (Ghasemkhani et al., 2018; Waters-Fuller, 1995) measured by the daily traffic congestion index of destination city j on day t from the Wind database.

The descriptive statistics of the main variables for the treatment and control groups are shown in Table B1 in Appendix B, respectively.¹⁰

4 | EMPIRICAL MODEL AND ESTIMATION

We employ a staggered DiD approach (Callaway & Sant'Anna, 2021) to robustly estimate the relationship between environmental uncertainty and delivery

performance in different phases, using Typhoon Lekima as the shock. This methodological framework is particularly suited to our study for several reasons.

Multiple phases of the shock. The phases of Typhoon Lekima serve as a suitable proxy for environmental uncertainty because each phase of the typhoon presents a distinct level of unpredictability and potential disruption.

Staggered nature of the shock. Typhoon Lekima did not affect all of China but was limited to specific cities like Hangzhou, Taizhou, Qingdao, Wuxi, and so forth. In addition, the cities were affected by Lekima at different times.

Unanticipated nature of the shock. The impact of Typhoon Lekima was largely unanticipated, with forecasts failing to accurately predict its trajectory and impact. For example, the inaccurate predictions of Lekima's landfall in cities like Ningbo, Jiaxing, and Shanghai, along with the unexpected impact on Lianyungang and Shaoxing, highlight the exogenous nature of this event.

In employing this staggered DiD approach, we leverage the differential timing of Typhoon Lekima's impact across various cities to robustly estimate the causal effect of environmental uncertainty on delivery performance. This methodology effectively controls for unobserved time-invariant factors and captures the dynamic response of delivery performance to the environmental shock (Fich et al., 2023).

4.1 | Environmental uncertainty and delivery performance

Our research examines the impact of environmental uncertainty on delivery performance, employing a staggered DiD. The model is formulated as follows:

$$\begin{aligned} \text{Promised}_{ijt} = & \beta_1 \text{TreatForecast}_{ijt} + \beta_2 \text{TreatActive}_{ijt} \\ & + \beta_3 \text{TreatRecovery}_{ijt} + \delta X + \eta_{ij} + \lambda_t + \varepsilon_{ijt}. \end{aligned} \quad (1)$$

As reflected in the model, we account for Typhoon Lekima's impact on the delivery performance of the affected cities across different phases (forecast, active, and recovery). Before the empirical analysis, we conduct a thorough examination to validate the parallel trends assumption for our staggered DiD approach (Fich et al., 2023; He & Tian, 2013; Kistler et al., 2021). This involves analyzing the growth patterns in delivery performance metrics for both the treatment and control groups to ensure they are comparable. Our findings, detailed in

Table C1 of Appendix C, demonstrate similar trends in both groups prior to the onset of Typhoon Lekima. This similarity supports the validity of the parallel trends assumption in our study.

Columns (I) and (II) in Table 2 present the estimates for the staggered DiD model in Equation (1). The coefficients of *TreatForecast* ($\beta_1 = -.265$, $p = .000$ for *PromisedOn*; $\beta_1 = -.267$, $p = .000$ for *PromisedBy*) and *TreatActive* ($\beta_2 = -.077$, $p = .000$ for *PromisedOn*; $\beta_2 = -.075$, $p = .000$ for *PromisedBy*) are significantly negative. These results indicate a decline in delivery performance during the forecast and active phases of Lekima when compared to the pre-disaster phase (i.e., the reference group in our dummy coding).

In addition, two interesting findings can be discovered from Columns (I) and (II) in Table 2. First, the impact during the active phase (*TreatActive*) is less pronounced than during the forecast phase (*TreatForecast*) for both *PromisedOn* and *PromisedBy*. Second, the results show a better delivery performance during the recovery phase compared to the pre-disaster phase, as indicated by *TreatRecovery* ($\beta_3 = .004$, $p = .028$ for *PromisedOn*; $\beta_3 = .004$, $p = .014$ for *PromisedBy*). We further discuss the intuition behind these two findings below.

First, contrary to initial expectations, the on-time delivery performance during the active phase of Lekima was not as negatively impacted as in the forecast phase. This outcome, which may seem counter-intuitive at first, can be attributed to RiRiShun's adoption of a wait-and-see policy in response to the forecast announcement of Typhoon Lekima. This strategy involved a reactive approach, closely monitoring the evolving conditions without making immediate operational changes. The primary focus was on assessing weather forecasts and updates, maintaining the operational status quo until more definitive information about the typhoon's impact was available.

This wait-and-see policy allowed for flexibility and adaptability in decision-making, with actions based on the latest information. However, it also carried the risk of a delayed response, potentially leaving the organization unprepared for rapid changes in the situation. Despite this risk, one key advantage was the conservation of resources, as the organization avoided committing resources prematurely, thus preventing unnecessary expenditures or logistical adjustments.

During the forecast phase, RiRiShun adopted a measured approach, opting not to adjust its infrastructure capacity immediately. Instead, RiRiShun maintained regular operations while closely monitoring the evolving situation. Hence, in this forecast stage, inbound items continued as usual, while outbound movements were

TABLE 2 The impact of environmental uncertainty on delivery performance.

	Full-sample			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(I)	(II)	(III)	(IV)
<i>Distance</i>	−0.014*** (0.000)	−0.010*** (0.000)	−0.041*** (0.002)	−0.028*** (0.002)
<i>ProductQuantity</i>	−0.003* (0.001)	−0.003** (0.001)	−0.003* (0.001)	−0.003* (0.001)
<i>LastMileHub</i>	0.741*** (0.017)	0.544*** (0.021)	0.743*** (0.017)	0.547*** (0.022)
<i>Congestion</i>	−0.029*** (0.003)	−0.028*** (0.003)	−0.024*** (0.003)	−0.022*** (0.003)
<i>TreatForecast</i>	−0.265*** (0.007)	−0.267*** (0.007)	−0.113*** (0.006)	−0.114*** (0.006)
<i>TreatActive</i>	−0.077*** (0.005)	−0.075*** (0.005)	−0.025*** (0.005)	−0.024*** (0.004)
<i>TreatRecovery</i>	0.004* (0.002)	0.004* (0.002)	−0.004* (0.002)	−0.005* (0.002)
<i>HighComplex</i>			0.167*** (0.013)	0.113*** (0.012)
<i>TreatForecast</i> × <i>HighComplex</i>			−0.362*** (0.013)	−0.365*** (0.013)
<i>TreatActive</i> × <i>HighComplex</i>			−0.135*** (0.012)	−0.134*** (0.012)
<i>TreatRecovery</i> × <i>HighComplex</i>			0.022*** (0.004)	0.022*** (0.004)
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes
Observations	458,357	458,357	458,357	458,357
<i>F</i>	798.8	520.5	563.0	391.5

Note: Standard errors clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

slowed or put on hold until the typhoon's impact became clear. This approach led to delays in deliveries during the forecast stage compared to the pre-disaster stage. To validate this observation, we conducted an event study (see Appendix D) and showed the accumulation of abnormal orders during the typhoon. This analysis reveals a significant bunching of orders during the forecast phase, a direct consequence of the wait-and-see policy. Subsequently, during the active Lekima phase, the number of abnormal orders decreased compared to the forecast phase. This decrease suggests that once the typhoon's effects became clear, RiRiShun was able to more effectively allocate its resources and manage the backlog of orders from the forecast phase.

Another interesting observation is that delivery performance in the recovery period is better when compared to the pre-Lekima period for the treatment group. We will further investigate the explanations for this finding in Section 4.2.

4.2 | Comparing low-complexity and high-complexity supply chains

In this section, we further investigate the differential impacts of environmental uncertainty phases on delivery performance between low-complexity and high-complexity supply chains. The model is formulated as follows:

TABLE 3 The impact of environmental uncertainty by supply chain complexity.

	Low-complexity supply chains		High-complexity supply chains	
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(I)	(II)	(III)	(IV)
<i>Distance</i>			−0.034*** (0.003)	−0.031*** (0.003)
<i>ProductQuantity</i>	−0.005** (0.002)	−0.005** (0.002)	−0.001 (0.003)	−0.000 (0.002)
<i>LastMileHub</i>	0.760*** (0.027)	0.557*** (0.033)	0.715*** (0.032)	0.456*** (0.039)
<i>Congestion</i>	−0.014*** (0.003)	−0.014*** (0.003)	−0.030*** (0.005)	−0.030*** (0.005)
<i>TreatForecast</i>	−0.102*** (0.006)	−0.104*** (0.006)	−0.491*** (0.012)	−0.494*** (0.012)
<i>TreatActive</i>	−0.017*** (0.005)	−0.017*** (0.004)	−0.169*** (0.011)	−0.167*** (0.011)
<i>TreatRecovery</i>	0.001 (0.002)	−0.000 (0.002)	0.010** (0.004)	0.011** (0.003)
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes
Observations	244,530	244,530	213,827	213,827
<i>F</i>	201.1	108.2	371.7	325.2

Note: The coefficient on *Distance* is omitted because *Distance* takes the value of zero for low-complexity supply chains (within-city deliveries). Standard errors clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

$$\begin{aligned}
 \text{Promised}_{ijt} = & \beta_1 \left(\text{TreatForecast}_{ijt} \times \text{HighComplex}_{ijt} \right) \\
 & + \beta_2 \left(\text{TreatActive}_{ijt} \times \text{HighComplex}_{ijt} \right) \\
 & + \beta_3 \left(\text{TreatRecovery}_{ijt} \times \text{HighComplex}_{ijt} \right) \\
 & + \beta_4 \text{TreatForecast}_{ijt} + \beta_5 \text{TreatActive}_{ijt} \\
 & + \beta_6 \text{TreatRecovery}_{ijt} + \beta_7 \text{HighComplex}_{ijt} \\
 & + \delta X + \eta_{ij} + \lambda_t + \varepsilon_{ijt}.
 \end{aligned} \tag{2}$$

The analysis of Equation (2) is presented in Columns (III) and (IV) in Table 2, reveals that the estimated coefficients ($\text{TreatForecast} \times \text{HighComplex}$, $\text{TreatActive} \times \text{HighComplex}$) are significantly negative for both the forecast and active phases of Lekima. This is observed for both *PromisedOn* and *PromisedBy* metrics. These findings suggest that high-complexity supply chains experience a more pronounced negative impact from environmental uncertainty during the forecast and active phases compared to low-complexity supply chains.

Contrary to our expectations, high-complexity supply chains perform better in delivery performance than low-complexity supply chains during the recovery phase. This is evident from the positive coefficients for *PromisedOn* ($\beta_3 = .022$, $p = .000$) and *PromisedBy* ($\beta_3 = .022$, $p = .000$).

As the impact of the exogenous shock from Typhoon Lekima began to wane, conventional expectations would suggest that delivery performance in the recovery phase would revert to levels comparable to those before the typhoon (pre-disaster levels). However, our analysis in Table 3 reveals a different scenario, particularly for high-complexity supply chains. While delivery performance in low-complexity supply chains returned to pre-disaster levels during the recovery period, high-complexity supply chains exhibited a surprising enhancement, outperforming their pre-disaster delivery performance.

This notable observation prompted us to consider the possibility that the firm utilized the crisis as an opportunity for sustained structural improvements in its delivery performance standards. To investigate this, we focused on the actual fulfillment times. Our analysis, detailed in Table 4, reveals a progressive decrease in the coefficients

TABLE 4 Effects on extending recovery-Lekima period.

	High-complexity supply chains		
	FTime	FTime	FTime
	(I)	(II)	(III)
Distance	27.496*** (0.672)	27.723*** (0.609)	27.568*** (0.593)
ProductQuantity	1.510** (0.483)	1.106* (0.471)	1.001* (0.430)
LastMileHub	−39.532*** (6.867)	−40.419*** (6.237)	−41.132*** (6.034)
Congestion	6.852*** (1.308)	11.488*** (1.320)	10.332*** (1.280)
TreatForecast	35.672*** (1.576)	35.098*** (1.540)	35.023*** (1.535)
TreatActive	9.340*** (1.409)	9.327*** (1.364)	9.296*** (1.347)
TreatRecovery (1-week)	−2.577** (0.900)		
TreatRecovery (2-week)		−2.968*** (0.836)	
TreatRecovery (3-week)			−3.258*** (0.808)
SKU-city fixed effects	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes
Observations	213,827	268,350	281,925
F	339.2	410.6	425.1

Note: Standard errors clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

for *TreatForecast*, *TreatActive*, and *TreatRecovery*. This indicates a consistent reduction in fulfillment times, extending from the forecast phase right through to the recovery phase. Notably, this trend of improvement in delivery performance continued even when we expanded our analysis to cover a 2-week or 3-week period post-disaster in the recovery phase. This finding indicates a more substantial operational improvement within high-complexity supply chains, highlighting the potential for firms to transform challenges posed by disruptions into opportunities for lasting operational enhancements.

4.3 | Analyzing the impact of response mechanisms

In this section, we examine the effectiveness of two response mechanisms—adjusting throughput efficiencies

at destination centers and implementing transfer centers—within high-complexity supply chains. Our analysis reveals that both mechanisms successfully mitigate the negative impact on delivery performance during the typhoon's forecast and active phases. However, there is no evidence that they have the same effect during the recovery phase.

We begin with throughput efficiency at destinations centers as a response mechanism to mitigate the adverse effects of environmental uncertainty on delivery performance. A higher throughput efficiency signifies a greater flow at destination center. Our analysis, presented in Columns (I) and (II) of Table 5, reveals that an increase in throughput efficiency by one standard deviation leads to a notable improvement in delivery performance. Specifically, during the forecast period of Lekima (*TreatForecast* $\times$ *TE*), we observe a 1.5% increase in *PromisedOn* and a 1.4% increase in *PromisedBy* performance. This trend continues into the active phase, with a 1.1% enhancement in both performance metrics (*TreatActive* $\times$ *TE*).

Next, we assess the impact of utilizing transfer centers as a strategic response to improve delivery performance. Consistent with existing literature (e.g., Huang et al., 2016), which suggests performance gains through acquiring reactive capacity during demand surges, we analyze the data presented in Columns (III) and (IV) of Table 5. The results indicate a substantial improvement in delivery performance during the forecast period (*TreatForecast* $\times$ *Transfer*), with increases of 53.7% for *PromisedOn* and 49.9% for *PromisedBy*. In the active phase, the continued use of transfer centers further enhances delivery performance by 15.8% for *PromisedOn* and 18.3% for *PromisedBy* (*TreatActive* $\times$ *Transfer*).

In our final analysis, presented in Columns (V) and (VI) of Table 5, we investigate whether throughput efficiency adjustments at destination centers and implementations of transfer centers act as substitutable or complementary mechanisms in terms of delivery performance. Interestingly, the concurrent execution of both mechanisms during the forecast phase (*TreatForecast* $\times$ *TE* $\times$ *Transfer*) has a counterproductive effect, reducing delivery performance by 1.5% for *PromisedOn* and 1.3% for *PromisedBy*.

4.4 | Robustness checks

By comparing changes in delivery performance before and after the typhoon in affected and unaffected areas, we can isolate the effect of this environmental shock from other concurrent influences. Interested readers may find value in exploring our extensive robustness checks in the appendices. These checks include using air pollution as an

TABLE 5 The impact of throughput efficiency and transfer center.

	High-complexity supply chains					
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(I)	(II)	(III)	(IV)	(V)	(VI)
<i>TreatForecast</i> × <i>TE</i>	0.015*** (0.002)	0.014*** (0.002)			0.014*** (0.002)	0.013*** (0.002)
<i>TreatActive</i> × <i>TE</i>	0.011*** (0.003)	0.011*** (0.003)			0.012*** (0.003)	0.011*** (0.003)
<i>TreatRecovery</i> × <i>TE</i>	−0.001 (0.001)	−0.001 (0.001)			−0.000 (0.001)	−0.000 (0.001)
<i>TreatForecast</i> × <i>Transfer</i>			0.537*** (0.056)	0.499*** (0.058)	0.555*** (0.064)	0.502*** (0.074)
<i>TreatActive</i> × <i>Transfer</i>			0.158* (0.072)	0.183** (0.060)	0.202* (0.088)	0.191* (0.081)
<i>TreatRecovery</i> × <i>Transfer</i>			−0.007 (0.027)	−0.024 (0.026)	0.039 (0.028)	0.025 (0.028)
<i>TreatForecast</i> × <i>TE</i> × <i>Transfer</i>					−0.015** (0.005)	−0.013** (0.005)
<i>TreatActive</i> × <i>TE</i> × <i>Transfer</i>					−0.015 (0.013)	−0.005 (0.008)
<i>TreatRecovery</i> × <i>TE</i> × <i>Transfer</i>					−0.004 (0.003)	−0.005 (0.003)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
SKU-city fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	213,827	213,827	213,827	213,827	213,827	213,827
<i>F</i>	256.8	222.1	244.9	213.8	151.7	131.0

Note: To conserve space, we omit the coefficients on control variables including *TreatForecast*, *TreatActive*, *TreatRecovery*, *TE*, *Transfer*, *Distance*, *LastMileHub*, *Congestion*, and *ProductQuantity*. Standard errors are clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

instrumental variable to address potential biases in ruling out active coordination effects (Appendix E). Additionally, we employ the creation of a placebo treatment group as a falsification test (Appendix F). We also address alternative biases, such as the possibility of coastal cities being more prone to typhoons and potential preferential treatment of certain products (Appendix G).

5 | DISCUSSION

In this section, we offer answers to the three research questions introduced earlier in Section 1. First, we discuss the relationship between environmental uncertainty and delivery performance. Second, we show that the performance of a high-complexity supply chain is different from

a low-complexity supply chain when facing environmental uncertainty. Third, we cover insights into two types of response mechanisms. The first mechanism focuses on throughput efficiencies at destination centers. The second mechanism focuses on channeling deliveries through transfer centers, effectively using them as reactive capacities. We further discuss whether these two response mechanisms are complementary or substitutable.

5.1 | Impact of environmental uncertainty on delivery performance: Various phases

Our analysis of the impact of environmental uncertainties on delivery performance, with a specific focus on

Typhoon Lekima, has unearthed several critical insights for supply chain professionals. Our study highlights the effects of different phases—forecast, active, and recovery—on delivery performance.

During the forecast phase of Typhoon Lekima, a common approach was the wait-and-see policy. While this cautious stance is understandable, it can lead to a significant backlog of orders, especially in high-complexity supply chains. The resultant uncertainty and delayed decision-making during this phase can be more disruptive than the typhoon itself, highlighting the importance of proactive planning and the adoption of flexible strategies in anticipation of environmental threats.

As the typhoon progressed to landfall, the affected orders had already been identified, allowing for a more targeted and efficient response. Interestingly, we found that the delivery performance during this active phase was less negatively impacted than during the forecast phase. This period demonstrates the resilience and adaptability of supply chains, effectively managing disruptions once the full extent of the impact becomes clear.

The recovery phase revealed an unexpected pattern. Delivery performance not only returned to pre-typhoon levels but actually showed an improvement. This indicates that the crisis might have acted as a catalyst for structural improvements or led to the adoption of more effective resource allocation strategies. The recovery phase thus highlighted the ability of supply chains to not only recover but also to evolve and enhance their operations in response to environmental challenges.

Based on these insights, we advise supply chain professionals to take a more proactive approach during the forecast phase to reduce the disruptive effects of uncertainty. In the active phase, it is crucial to maintain adaptability and resilience to manage immediate impacts efficiently. Post-typhoon, during the recovery phase, there lies a valuable opportunity to reassess and refine supply chain structures and processes, turning challenges into catalysts for improvements.

5.2 | Low-complexity supply chains versus high-complexity: Divergent impacts

We observed a significant divergence in how these two types of supply chains responded to the typhoon. High-complexity supply chains were more adversely affected during the forecast and active phases of the typhoon compared to low-complexity supply chains. This was evident in the notable negative impacts on both *PromisedOn* and *PromisedBy* delivery metrics, aligning with the heightened vulnerability of more complex supply chains to environmental uncertainties.

There is a surprising turnaround in the recovery phase. Contrary to expectations, the recovery phase presented a surprising turnaround for high-complexity supply chains. Instead of merely returning to pre-typhoon performance levels, these supply chains exhibited significant improvements, surpassing their pre-disaster delivery performance. This finding was not just unexpected but also challenged the conventional understanding of recoveries from disasters in supply chains.

Our investigation indirectly probes into the operational improvement beyond crisis management. Further analysis of this phenomenon, particularly focusing on fulfillment times, revealed a consistent improvement across all phases of the typhoon, extending even 3 weeks into the recovery period. Feedback from RiRiShun revealed that some of the temporary transportation nodes and links used during Typhoon Lekima became permanent features in their delivery routes. Initially, support came from ad hoc third-party transportation and logistics services using alternative routes during the active disaster phase. These services evolved into more permanent flexible contracts in the recovery period. This opportunity during the crisis allowed RiRiShun to build effective inventory prepositioning for rapid response. The newly optimized network established through these contracts post-disaster enhances the overall response capacity.

These insights are crucial for supply chain strategy, particularly for managing complex networks. They underscore the importance of agility and adaptability when facing environmental uncertainties. More significantly, they highlight the potential of crises to act as catalysts for strategic transformation and sustained efficiency gains. For supply chain professionals, this study serves as a compelling example of leveraging environmental challenges not just as obstacles to overcome but as opportunities for strategic growth and operational improvement.

5.3 | Response mechanisms in high-complexity supply chains

Our research highlights two key response mechanisms for firms to enhance their delivery performance in high-complexity supply chains.

5.3.1 | Throughput efficiency

By focusing on managing throughput efficiency at destination centers, RiRiShun can significantly speed up the transition of inbound orders to outbound, reducing idle times. To achieve this, RiRiShun leverages various operational improvements such as refining scheduling

processes, enhancing the layout of operations for smoother flow, and assessing the need for automation to improve handling times and storage optimization. An operations analyst at RiRiShun highlighted that such efforts are fostered through collaborative planning teams among different parts of the organization. This can take the form of improving the accuracy of demand forecasting and ensuring tighter coordination of inventory replenishment between the upstream production and logistics teams.

To segregate and quantify the specific factors contributing to the throughput efficiency of the destination centers are beyond this Technical Note. Nonetheless, our findings reveal that focusing on improving flow and efficiency at destination centers works for both forecast and active phases but more for the former. In such contexts, according to additional retrospective insights from the logistics manager, the RiRiShun as a logistics arm that traditionally supports a make-to-order upstream production system, demonstrates the capacity for strategic adaptation in response to disruptions. This adaptation involves a temporary shift, as disruption compels the system to resemble a make-to-stock more. Therefore, throughput efficiency becomes critically important in scrutinizing service levels at destination centers when supporting a make-to-stock system behavior rather than a make-to-order as explained further by the logistics manager.

5.3.2 | Transfer centers

By focusing on channeling deliveries through internal and external transfer centers, RiRiShun develops reactive capacities that enhance its ability to respond to disruptions. When a third-party transfer center is used, RiRiShun incurs a cost premium. Transfer centers play a significant role during the forecast and active phases of Lekima, aligning with the literature that advocates for reactive capacity acquisition during high-demand periods (Huang et al., 2016).

5.3.3 | Substitutability versus complementarity

The lead data manager at RiRiShun explains that their strategy involves using a combined approach to mitigate disruptions. Local teams at downstream destination centers strive to enhance throughput efficiency, for instance, by consolidating more orders into a single delivery period and scheduling additional trucks for delivery.

Concurrently, regional teams employ transfer centers as preventive measures, rerouting packages to manage potential disruptions.

Our analysis highlights a critical dynamic in the concurrent execution of these response strategies during the forecast phase of disruptions. We found that simultaneously activating additional transfer centers along the supply chain while adjusting throughput efficiency adjustments at the destination centers can detrimentally impact delivery performance. This indicates that while each strategy is effective independently, their combined application can be counterproductive. This finding implies the importance of accurately selecting the most appropriate initial response mechanism at the outset. For instance, if local destination teams pursue throughput efficiency adjustments for cost efficiency, but regional teams concurrently activate transfer centers anticipating higher demand, the efforts and resources dedicated to throughput adjustments may become redundant, as the transfer centers already mitigate the anticipated delivery challenges.

Regarding the tradeoff between leveraging an additional transfer center and locally optimizing throughput efficiency at the destination center, we recognize that utilizing additional reactive capacities through a transfer center—albeit internal or a third-party service supporting package redirection or temporary holding—may seem inefficient during the forecast period. Nonetheless, destination centers can better optimize the throughput of incoming and outgoing orders upon completion of the transfer center plan. This approach allows for a more measured and sequential response, rather than simultaneously executing both mechanisms, which would be less effective for promised delivery times.

5.4 | Concluding remarks

Our study, centered on the disruptions caused by Typhoon Lekima, offers essential insights into managing supply chain disruptions amidst environmental uncertainty. By exploiting the typhoon's lifecycle, we provide a detailed framework for addressing similar disruptive events in supply chain management. We believe these insights highlight the broader applicability of managing demand uncertainty phenomena. We encourage further exploration into how different uncertainties, both natural and market-driven, impact supply chains. Future research could focus on specific strategies for mitigating these uncertainties, examining their effects on various supply chain structures, and exploring innovative response mechanisms to enhance resilience.

OPEN RESEARCH BADGES



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ENDNOTES

- ¹ Interested readers may refer to Angrist and Krueger (2001), Morretti (2011), and Lee et al. (2018) for studies that use severe weather conditions as exogenous shocks.
- ² To validate consistency with our use of the concept of supply chain complexity, we conduct semi-structured interviews, which is common in OM research (see Flynn et al., 1990). The interviews were conducted with the director of logistics for discussion on complexities in the strategic level decision-making of the inter-city network, and the operations manager that coordinates within-city vehicle routing and scheduling operations.
- ³ <http://www.bjnews.com.cn/news/2019/08/15/615677.html>.
- ⁴ <http://www.weather.com.cn>.
- ⁵ The route map is generated using the web application services of the Meteorological Bureau of Shenzhen Municipality (please see <http://218.17.86.92/en/index.html>). The different levels of typhoon impact strength are defined following Chinese National Standard—Grade of Tropical Cyclones (please see https://tcdata.typhoon.org.cn/data/doc/TC_std.pdf).
- ⁶ Data are available at <https://doi.org/10.1287/msom.2021.0994>.
- ⁷ <https://www.rrswl.com/>.
- ⁸ See <https://www.rrswl.com/ckwd/searchCangku?cangkLeix=0> for the warehouse inquiry page, where we use the map API (99 centers pinned with addresses) to create our distance matrix.
- ⁹ <https://content.parcel solutions.net/ba692e52-45bd-427a-90ba-0abb c70684e2/general/late%20delivery%20credit%20scale%20eguide.pdf>.
- ¹⁰ For ease of readability, the subscripts of variables in the text, tables and figures are omitted from hereon.
- ¹¹ Particulate matter (PM) includes microscopic matter suspended in air. PM₁₀ refers to particulate matter smaller than 10 g/m³ in diameter.
- ¹² Stock et al. (2002) develop benchmarks for the weak instrument test: when the number of instruments is 1, 2, 3, 5, and 10, the suggested thresholds of *F*-statistics are 8.96, 11.59, 12.83, 15.09, and 20.88, respectively.

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How to cite this article: Ignatius, J., Li, R., Yin, Y., & Jia, K. (2024). Steering through the storm: Environmental uncertainty and delivery performance. *Journal of Operations Management*, 70(8), 1286–1309. <https://doi.org/10.1002/joom.1331>

APPENDIX A: IMPORTANCE OF PROMISED DELIVERY DATE DISPLAYED IN ORDER INFORMATION

See Figures A1 and A2.

amazon Checkout (1 item)

1 Shipping address Cecilia
3000 184TH ST SW
LYNNWOOD, WA 98037-4718
[Add delivery instructions](#) [Change](#)

2 Choose special delivery options

Shipping to: Cecilia, 3000 184TH ST SW, LYNNWOOD, WA, 98037-4718 United States [Change](#)

 **Danby DFF101B1BSLDB 10.1 Cu.Ft. Top Mount Freezer, Energy-Star Rated Apartment Refrigerator with Smudge Free Stainless Look**
\$588.00 - Quantity: 1
Sold by: Amazon.com Services LLC

What we are doing during COVID-19. [Learn More](#)

Delivery Method

☐ Front Porch Details >
FREE Shipping

☒ **FREE, Inside Entryway**
As soon as **Thursday, Oct 27**
Scheduled delivery to the first dry area inside your residence. [Details](#) >

☐ **\$19.99, Room of Choice Delivery**
As soon as **Thursday, Oct 27**
Scheduled delivery to your room of choice inside your residence. Does not include unpack or assembly. [Details](#) >

Choose your preferred delivery day

October

SUN	MON	TUE	WED	THU	FRI	SAT
23	24	25	26	27	28	29
NOV						
30	31	1	2	3	4	5

[Show more](#) >

Choose your delivery time

6AM - 9AM 9AM - 12PM 12PM - 3PM

FIGURE A1 Screenshot of an order on Amazon.

JD 京东自营

T型门冰箱热卖榜·第1名

海尔 (Haier) 465升星辉系列一级能效...
【全新升级款】465L星辉节能大师

¥3799
¥3899.00

支持7天无理由退货 跨自营/店铺促销

[赠品] 冰箱延保3年+清洗服务 x1
赠品数量有限, 先到先得

服务 可选增值服务 >

配送 京东快递(送货上门) >
Promised Delivery Date 送货: 9月24日 [周六]
安装: 9月24日 [周六]

商品金额 ¥3899.00
运费 ¥0.00
优惠券 无可用 >
以旧换新 ⓘ 最高优惠 ¥420.00 >
已隐藏本单不可使用的虚拟资产 ~

¥3899.00 提交订单

FIGURE A2 Screenshot of an order on JD.com.

APPENDIX B: DESCRIPTIVE STATISTICS

	All		Treatment group		Control group	
	Mean	SD	Mean	SD	Mean	SD
<i>PromisedOn</i>	0.940	0.220	0.930	0.240	0.942	0.218
<i>PromisedBy</i>	0.951	0.201	0.938	0.227	0.953	0.197
<i>FTime</i>	69.990	57.539	58.953	54.908	71.511	57.726
<i>TE</i>	8.967	43.527	8.578	27.868	9.021	45.268
<i>Transfer</i>	0.042	0.196	0.018	0.128	0.045	0.203
<i>Distance</i>	2.716	2.948	2.409	2.752	2.758	2.972
<i>ProductQuantity</i>	1.349	0.745	1.366	0.743	1.347	0.745
<i>LastMileHub</i>	0.999	0.037	0.998	0.049	0.999	0.035
<i>Congestion</i>	1.519	0.159	1.468	0.140	1.525	0.161
Observations	458,357		55,551		402,806	

Note: The units of *PromisedOn*, *PromisedBy*, *TE*, *Transfer*, and *LastMileHub* are Percentage. The unit of *Distance* is log(Kilometer). The unit of *FTime* is Hour. The unit of *ProductQuantity* is Count. The unit of *Congestion* is Index.

TABLE B1 Descriptive statistics.

APPENDIX C: INVESTIGATIONS ON PARALLEL TRENDS

TABLE C1 Parallel trends: Dynamic coefficient test.

	Full-sample		Low-complexity supply chains		High-complexity supply chains	
	PromisedOn Growth	PromisedBy Growth	PromisedOn Growth	PromisedBy Growth	PromisedOn Growth	PromisedBy Growth
	(I)	(II)	(III)	(IV)	(V)	(VI)
$Lekima^{(t-10)}$	0.003 (0.010)	0.004 (0.010)	0.001 (0.015)	0.003 (0.014)	0.009 (0.015)	0.011 (0.014)
$Lekima^{(t-9)}$	0.009 (0.011)	0.012 (0.011)	0.001 (0.016)	0.007 (0.016)	0.012 (0.015)	0.014 (0.014)
$Lekima^{(t-8)}$	0.006 (0.010)	0.009 (0.010)	-0.005 (0.016)	-0.006 (0.015)	0.016 (0.014)	0.020 (0.013)
$Lekima^{(t-7)}$	0.003 (0.011)	0.005 (0.010)	0.003 (0.016)	0.001 (0.015)	0.007 (0.014)	0.012 (0.014)
$Lekima^{(t-6)}$	0.005 (0.011)	0.005 (0.010)	-0.002 (0.016)	-0.002 (0.015)	0.011 (0.014)	0.012 (0.013)
$Lekima^{(t-5)}$	0.004 (0.011)	0.002 (0.011)	-0.006 (0.016)	-0.006 (0.015)	0.007 (0.015)	0.005 (0.014)
$Lekima^{(t-4)}$	0.005 (0.011)	0.005 (0.011)	0.003 (0.016)	0.005 (0.016)	0.003 (0.015)	0.002 (0.015)
$Lekima^{(t-3)}$	0.000 (0.012)	0.001 (0.011)	-0.009 (0.017)	-0.005 (0.017)	0.011 (0.016)	0.010 (0.015)
$Lekima^{(t-2)}$	0.006 (0.011)	0.008 (0.011)	0.009 (0.016)	0.013 (0.016)	0.004 (0.015)	0.001 (0.014)
$Lekima^{(t+0)}$	-0.055*** (0.011)	-0.054*** (0.010)	-0.040*** (0.015)	-0.040*** (0.014)	-0.076*** (0.015)	-0.076*** (0.014)
$Lekima^{(t+1)}$	-0.109*** (0.011)	-0.112*** (0.011)	0.015 (0.016)	0.011 (0.016)	-0.275*** (0.015)	-0.276*** (0.014)
$Lekima^{(t+2)}$	0.016 (0.012)	0.015 (0.012)	-0.073*** (0.017)	-0.072*** (0.016)	0.257*** (0.018)	0.256*** (0.017)
$Lekima^{(t+3)}$	0.013 (0.008)	0.014 (0.008)	0.009 (0.012)	0.008 (0.011)	0.022* (0.011)	0.025* (0.010)
Observations	331,359	334,063	187,633	189,256	143,726	144,807
F	20.32	23.24	4.16	4.36	80.35	90.84

Note: PromisedOnGrowth (PromisedByGrowth) is the daily growth rate of PromisedOn (PromisedBy) at the SKU-city level, that is, $PromisedOnGrowth_{ijt} = (PromisedOn_{ijt} - PromisedOn_{ijt-1}) / PromisedOn_{ijt-1}$ and $PromisedByGrowth_{ijt} = (PromisedBy_{ijt} - PromisedBy_{ijt-1}) / PromisedBy_{ijt-1}$. $Lekima^{(t-(+)^i)}$ is a dummy equal to one if the observation is the i th day before (after) the date the Lekima's forecast is announced and zero otherwise. $Lekima^{(t-1)}$ is the omitted benchmark group. Standard errors clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

APPENDIX D: THE ABNORMAL ORDERS DIFFERENTIALS BETWEEN ACTIVE- AND FORECAST-LEKIMA PHASES

In our analysis, we utilized a regression model to estimate the expected number of daily orders in affected cities during both the forecast and active stages of Typhoon Lekima, based on data from the pre-Lekima stage. This model allowed us to calculate the abnormal orders, which represent the discrepancy between the actual and predicted number of orders. We then determined the abnormal orders differential by subtracting the number of abnormal orders in the forecast stage from those in the active stage of Lekima. The results, presented in Table D1, reveal a significantly higher accumulation of orders during the forecast stage compared to the active stage, as evidenced by the *t*-test for abnormal orders differential.

APPENDIX E: ROBUSTNESS CHECKS: INSTRUMENTAL VARIABLE

In our base specification, identification of the impact of environmental uncertainty on delivery performance assumes that deliveries are uncorrelated with the regression errors, after controlling for the fixed effects and other control variables. This means that deliveries in the treatment and control groups are independently autonomous, and there is no active coordination to sacrifice delivery performance in the latter for the former during an exogenous event. For example, if the central planner allocates more resources for deliveries in the treated areas in anticipation of the impact of Lekima when the forecasts have been announced, this would then be a potential source of endogeneity. If this scenario occurs, we would have underestimated the impact of environmental uncertainty on delivery performance in Equation (1). We address this potential endogeneity issue by adopting the PM10,¹¹ which is widely used as an air pollution indicator (see Herrnstadt et al., 2021), as our instrumental variable, *AirPollution*.

We do so by exploiting the variation of air pollutants across cities, which strongly follows the wind direction (Herrnstadt et al., 2021). Given wind direction affects the travel pathway of typhoon Lekima, the instrument relevance condition is satisfied because cities in the treatment group (those impacted by Lekima) expects a decline in air pollutants relative to cities in the control group. Since there is exogeneity in air pollutants across cities and highly improbable that air pollution impacts delivery performance directly, *AirPollution* fulfills the exclusion criterion of a valid instrumental variable.

We have conducted the first stage regression of two-stage least squares (2SLS), where the Cragg-Donald Wald

TABLE E1 Endogeneity issue: 2SLS estimates of the second stage.

	Full-sample	
	PromisedOn (I)	PromisedBy (II)
<i>Distance</i>	−0.015*** (0.000)	−0.011*** (0.000)
<i>ProductQuantity</i>	−0.003** (0.001)	−0.004*** (0.001)
<i>LastMileHub</i>	0.737*** (0.011)	0.557*** (0.010)
<i>Congestion</i>	−0.017*** (0.003)	−0.016*** (0.003)
<i>TreatForecast</i>	−0.188*** (0.006)	−0.192*** (0.006)
<i>TreatActive</i>	−0.091*** (0.009)	−0.092*** (0.008)
<i>TreatRecovery</i>	0.054*** (0.006)	0.049*** (0.005)
SKU-city fixed effects	Yes	Yes
Day fixed effects	Yes	Yes
Observations	380,058	380,058
<i>F</i>	1165	891.9

Note: We use *AirPollution* as the instrumental variable.

****p* < .001; ***p* < .01; **p* < .05.

TABLE D1 The abnormal orders differentials between active- and forecast-Lekima phases.

	H0: Mean difference = 0			95% confidence interval	
	Mean difference	<i>t</i> -statistic	<i>p</i> -value	Lower	Upper
Abnormal orders differential	−844.916	10.309	0.000	−865.122	−824.710

Note: *N* = 16,270.

F -statistic is 25,915.31,¹² suggesting that the instrumental variable does not suffer from weak instrument issues. Table E1 reports the 2SLS estimates from the second stage, which matches the results of our base specification in Equation (1), validating our findings.

APPENDIX F: ROBUSTNESS CHECKS: FALSIFICATION TEST

To lend additional support to the validity of our results, we perform a falsification test for our staggered DiD analysis to check whether the typhoon event-induced delivery shocks are picking up some spurious effects. Following Nian et al. (2021) and Chetty et al. (2009), we perform a placebo test by randomly assigning treatment cities to create a pseudo or placebo treatment. Using the pseudo treatment, we rerun our regression models in Equation (1). We replicate this procedure 100 times, store the coefficient estimates and plot the distributions of placebo estimates in Figure F1. The estimated results after randomization are distributed centrally near zero and in line with normal distribution, indicating that the pseudo-treatment group has no significant effect on delivery performance. This placebo test provides support to our earlier results and rules out the possibility that our estimates have picked up some spurious effects.

APPENDIX G: ROBUSTNESS CHECKS: RULING OUT POTENTIAL BIASES

In our study, we take additional steps to address two potential sources of bias that could influence our findings.

First, we consider the possibility that the observed significant negative impact of uncertainty on performance might be influenced by the volume of deliveries, particularly since coastal cities, more prone to typhoons, also tend to have higher delivery volumes compared to inland cities. To address this concern, we focus our analysis on the coastal subsample and test our baseline specifications. The results, detailed in Table G1, align with our initial findings presented in Tables 2 and 5, thereby mitigating concerns related to delivery size bias.

Second, we explore the potential bias arising from preferential treatment of certain brands in deliveries. Given that RiRiShun is part of the Haier group but also services other brands, there could be a concern regarding priority delivery for Haier's products. To investigate this, we match products based on dimensions from Haier's 2019 product catalog with the SKUs in our database, identifying a significant number of refrigerators, washing machines, and televisions. After excluding these matched products, we rerun our baseline analysis. The consistency of these results, as shown in Table G2, with those in Tables 2 and 5, suggests that our findings are not skewed by any preferential treatment of Haier's products.

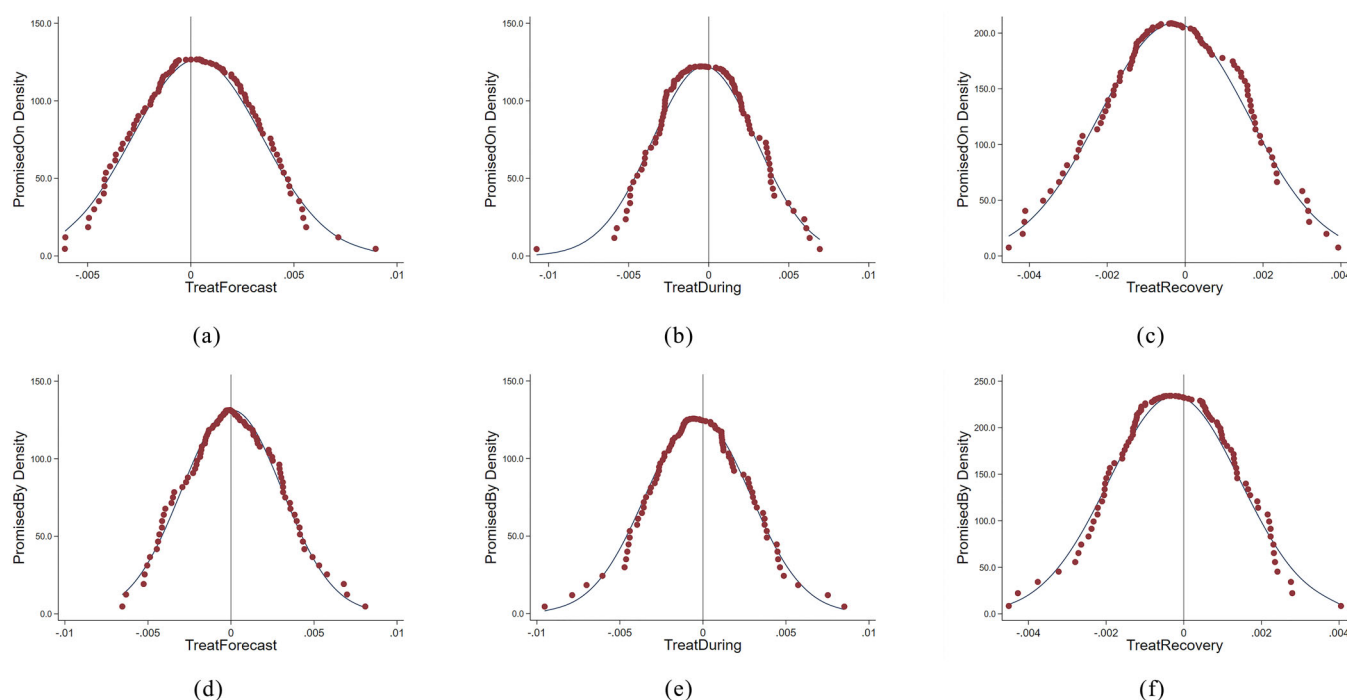


FIGURE F1 Distribution of placebo estimates. (a–c) Plot the distributions of estimates of pseudo effects on *PromisedOn*. (d–f) Plot the distributions of estimates of pseudo effects on *PromisedBy*.

TABLE G1 Robustness checks on coastal-city sample.

Panel A. Main effects				
	Full-sample			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(I)	(II)	(III)	(IV)
<i>TreatForecast</i>	−0.265*** (0.007)	−0.267*** (0.007)	−0.115*** (0.006)	−0.115*** (0.006)
<i>TreatActive</i>	−0.076*** (0.005)	−0.075*** (0.005)	−0.026*** (0.005)	−0.026*** (0.004)
<i>TreatRecovery</i>	0.004* (0.002)	0.004* (0.002)	−0.005* (0.002)	−0.006** (0.002)
<i>HighComplex</i>			0.226*** (0.019)	0.162*** (0.017)
<i>TreatForecast</i> × <i>HighComplex</i>			−0.359*** (0.013)	−0.362*** (0.013)
<i>TreatActive</i> × <i>HighComplex</i>			−0.133*** (0.012)	−0.130*** (0.012)
<i>TreatRecovery</i> × <i>HighComplex</i>			0.024*** (0.004)	0.025*** (0.004)
Other controls	Yes	Yes	Yes	Yes
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes
Observations	216,813	216,813	216,813	216,813
<i>F</i>	551.9	418.9	404.1	325.3
Panel B. Response mechanisms				
	High-complexity supply chains			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(V)	(VI)	(VII)	(VIII)
<i>TreatForecast</i> × <i>TE</i>	0.015*** (0.002)	0.014*** (0.002)		
<i>TreatActive</i> × <i>TE</i>	0.011*** (0.003)	0.011*** (0.003)		
<i>TreatRecovery</i> × <i>TE</i>	−0.001 (0.001)	−0.001 (0.001)		
<i>TreatForecast</i> × <i>Transfer</i>			0.547*** (0.057)	0.508*** (0.058)
<i>TreatActive</i> × <i>Transfer</i>			0.165* (0.072)	0.192** (0.060)
<i>TreatRecovery</i> × <i>Transfer</i>			0.004 (0.027)	−0.015 (0.026)
Other controls	Yes	Yes	Yes	Yes
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes

TABLE G1 (Continued)

Panel B. Response mechanisms				
	High-complexity supply chains			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(V)	(VI)	(VII)	(VIII)
Observations	106,476	106,476	106,476	106,476
<i>F</i>	212.6	205.1	203.0	198.2

Note: Coastal sample include cities in coastal provinces including Guangdong, Fujian, Zhejiang, Shanghai (municipality), Jiangsu, Shandong, and Liaoning. To conserve space, we omit the coefficients on explanatory variables (*TreatForecast*, *TreatActive*, *TreatRecovery*, *TE*, and *Transfer*) in Panel B and control variables (*Distance*, *LastMileHub*, *Congestion*, and *ProductQuantity*) in both panels. Standard errors are clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

TABLE G2 Robustness checks by excluding Haier products.

Panel A. Main effects				
	Full-sample			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(I)	(II)	(III)	(IV)
<i>TreatForecast</i>	−0.271*** (0.007)	−0.272*** (0.007)	−0.114*** (0.006)	−0.115*** (0.006)
<i>TreatActive</i>	−0.076*** (0.005)	−0.074*** (0.005)	−0.023*** (0.005)	−0.022*** (0.004)
<i>TreatRecovery</i>	0.005** (0.002)	0.005** (0.002)	−0.004 (0.002)	−0.004* (0.002)
<i>HighComplex</i>			0.185*** (0.014)	0.124*** (0.013)
<i>TreatForecast</i> × <i>HighComplex</i>			−0.364*** (0.013)	−0.368*** (0.013)
<i>TreatActive</i> × <i>HighComplex</i>			−0.138*** (0.012)	−0.137*** (0.012)
<i>TreatRecovery</i> × <i>HighComplex</i>			0.024*** (0.004)	0.024*** (0.004)
Other controls	Yes	Yes	Yes	Yes
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes
Observations	428,045	428,045	428,045	428,045
<i>F</i>	767.7	512.5	541.6	384.4
Panel B. Response mechanisms				
	High-complexity supply chains			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(V)	(VI)	(VII)	(VIII)
<i>TreatForecast</i> × <i>TE</i>	0.016*** (0.002)	0.015*** (0.002)		
<i>TreatActive</i> × <i>TE</i>	0.014*** (0.003)	0.013*** (0.003)		
<i>TreatRecovery</i> × <i>TE</i>	−0.000 (0.001)	−0.001 (0.001)		
<i>TreatForecast</i> × <i>Transfer</i>			0.540*** (0.061)	0.498*** (0.063)
<i>TreatActive</i> × <i>Transfer</i>			0.183** (0.069)	0.195** (0.066)
<i>TreatRecovery</i> × <i>Transfer</i>			0.008 (0.027)	−0.012 (0.026)
Other controls	Yes	Yes	Yes	Yes
SKU-city fixed effects	Yes	Yes	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes

TABLE G2 (Continued)

Panel B. Response mechanisms				
	High-complexity supply chains			
	<i>PromisedOn</i>	<i>PromisedBy</i>	<i>PromisedOn</i>	<i>PromisedBy</i>
	(V)	(VI)	(VII)	(VIII)
Observations	199,983	199,983	199,983	199,983
<i>F</i>	247.6	216.7	235.2	207.0

Note: To conserve space, we omit the coefficients on explanatory variables (*TreatForecast*, *TreatActive*, *TreatRecovery*, *TE*, and *Transfer*) in Panel B and control variables (*Distance*, *LastMileHub*, *Congestion*, and *ProductQuantity*) in both panels. Standard errors are clustered at the SKU-city level in parentheses.

*** $p < .001$; ** $p < .01$; * $p < .05$.